



CSE-476: Introduction to Natural Language Processing
Quiz 2

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

- Q1. Which type of model directly learns the decision boundary between classes?
- A. Generative models model $P(x, y)$, while discriminative models model $P(y | x)$
 - B. Discriminative models learn the data distribution and use that knowledge for classification
 - C. Discriminative models require no training data
 - D. Generative models are less explainable
- Q2. Which technique is used to mitigate overfitting by penalizing large weights?
- A. Laplace smoothing
 - B. Regularization
 - C. TF-IDF
 - D. Softmax normalization
- Q3. What is the limitation of the perceptron algorithm?
- A. It cannot use vector representations
 - B. It only works for linearly separable data
 - C. It requires probabilistic outputs
 - D. It cannot be trained with gradient descent
- Q4. Why is Laplace smoothing important?
- A. It improves cosine similarity
 - B. It reduces dimensionality
 - C. It avoids zero probabilities for unseen words
 - D. It speeds up gradient descent

- Q5. What happens in stochastic gradient descent (SGD) compared to gradient descent (GD)?
- A. SGD updates weights using the entire dataset every step
 - B. SGD updates weights using one or a subset of training examples
 - C. GD converges faster because updates are noisier
 - D. SGD cannot be used with neural networks
- Q6. What is the core intuition behind word embeddings?
- A. Words are represented by their dictionary definitions
 - B. Words that appear in similar contexts have similar meanings
 - C. Words are clustered by part of speech
 - D. Words are ranked by frequency
- Q7. What is the primary purpose of the Softmax function?
- A. To penalize large model weights
 - B. To reduce feature dimensionality
 - C. To select the class with the largest raw score without normalization
 - D. To convert raw class scores into a valid probability distribution
- Q8. In the perceptron algorithm, when are the weights updated?
- A. After every training example
 - B. Only when the prediction is correct
 - C. Only when the prediction is incorrect
 - D. Only after one full pass over the data
- Q9. What is the main goal of TF-IDF weighting?
- A. Increase the weight of very common words
 - B. Represent word order
 - C. Convert sparse vectors into dense embeddings
 - D. Highlight words that are frequent in a document but rare across documents
- Q10. Why is cosine similarity preferred over dot product when comparing word vectors?
- A. It ignores vector length while calculating similarity
 - B. It increases the influence of high-frequency words
 - C. It converts similarity scores into probabilities
 - D. It reduces the dimensionality of word embeddings